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The peak/off-peak (PR3/PR5) operating state hours (off peak, midnight to 6 a.m.) are pro-rata apportioned by nightly time duration to achieve an overall power density indicator result. In this particular case of equal time duration the two part-night power density indicators are averaged to achieve an overall power density indicator result.

$$D_{\mathrm{P}} = (0.01458 + 0.01685)/2 = 0.01571\, W.lx^{-1}.m^{-2}.$$

F.2.5.2 Worked example of a lighting scheme with constant light output (CLO)

This example uses the same physical and lighting design parameters as in the example above.

If the designed lighting scheme for PR3 operation uses a luminaire wattage of 20 W to achieve subcategory conformance at the end of the maintenance period; and if the lumen depreciation at the end of the maintenance period is, say 0.3 (L70), with CLO the initial luminaire wattage will be 14 W (20 W × 0.7).

The luminaire wattage appreciation curve over the maintenance period is the converse of the lumen depreciation curve. If the luminaire is assumed to have a straight lumen depreciation curve, the average luminaire wattage over the maintenance period is **(20 + 14)/2 = 17 W**.

For PR5 operation with CLO the initial luminaire wattage will be 7 W (10 W × 0.7). The average luminaire wattage over the maintenance period is **(10 + 7)/2 = 8.5 W**.

Alternatively, use the calculated lumen depreciation figures from IESNA LM-80-15 and IESNA TM-21-11 extrapolation including luminaire *in situ* life calculations for the luminaire concerned.

The system power (P) calculation for the above PR3 example with CLO is—

$$P = \sum_{k=1}^{n_{\mathrm{lp}}} P_{\mathrm{k}} + P_{\mathrm{ad}} \qquad \text{F7}$$

where

P = total system power of the lighting scheme (W)

P_{k} = operational power of light point = (17 +2) = 19 W

P_{ad} = total operational power of system devices = 30 W

Therefore the system power $(P) = (20 \times 19) + (30) = 410\, W$.

The power density indicator calculation (D_{P}) for the scheme for PR3 with CLO operation is—

$$D_{\mathrm{P}} = \frac{P}{\sum_{i=1}^{n} \left(\bar{E}_{\mathrm{i}} \cdot A_{\mathrm{i}}\right)} \qquad \text{F8}$$

where

D_{P} = power density indicator, in watts per lux per square metre ($W.lx^{-1}.m^{-2}$)

P = system power = 410 W

$\bar{E}_{\mathrm{i}}$ = maintained average horizontal illuminance is 1.79 lx (say) (for subcategory PR3)

A_{i} = size of the sub-area (1000 × 18) = 18 000 m^2

n = number of sub-areas to be lit = 1

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